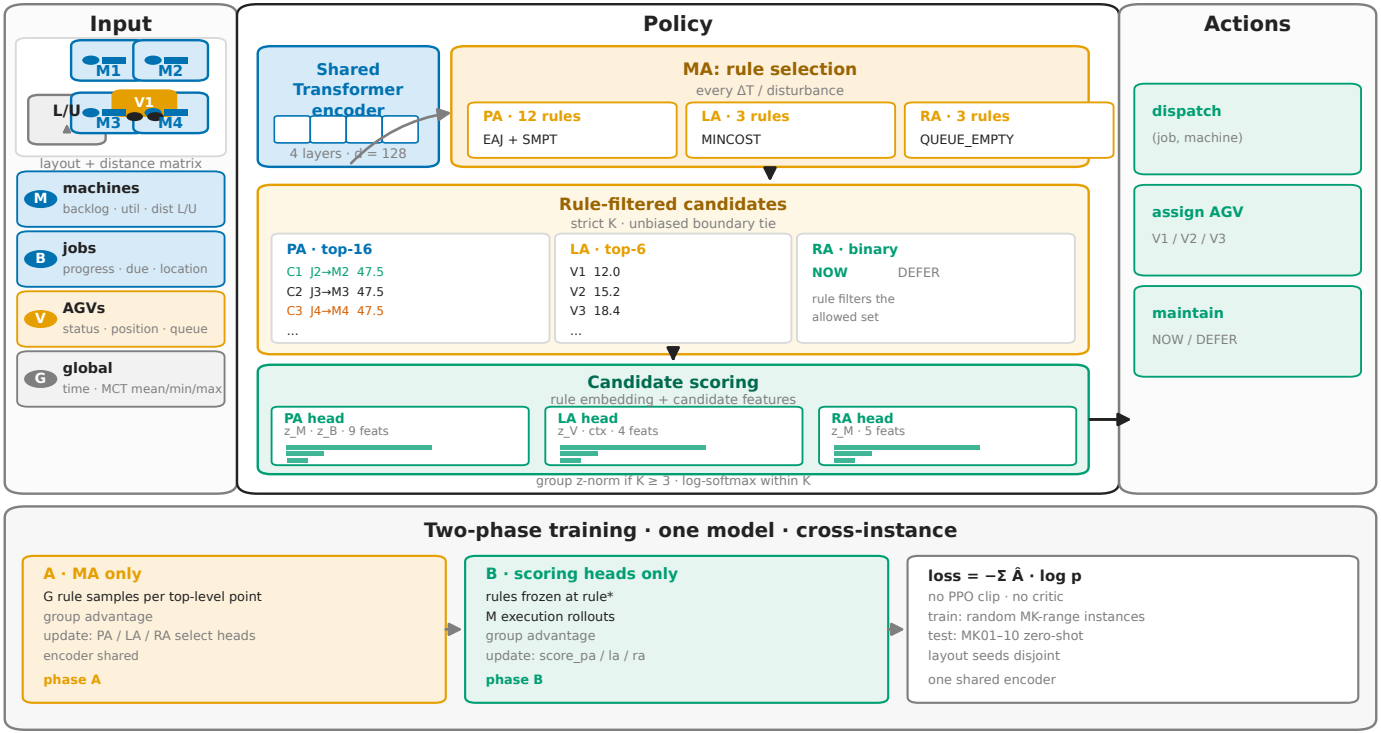

Input
M1
M2
L/U
V1
M3
M4
layout + distance matrix
M
machines
backlog · util · dist L/U
B
jobs
progress · due · location
V
AGVs
status · position · queue
G
global
time · MCT mean/min/max
Policy
Shared Transformer encoder
4 layers · d = 128
MA: rule selection
every ΔT / disturbance
PA · 12 rules
EAJ + SMPT
LA · 3 rules
MINCOST
RA · 3 rules
QUEUE_EMPTY
Rule-filtered candidates
strict K · unbiased boundary tie
PA · top-16
C1 J2→M2 47.5
C2 J3→M3 47.5
C3 J4→M4 47.5
...
LA · top-6
V1 12.0
V2 15.2
V3 18.4
...
RA · binary
NOW
DEFER
rule filters the allowed set
Candidate scoring
rule embedding + candidate features
PA head
z_M · z_B · 9 feats
LA head
z_V · ctx · 4 feats
RA head
z_M · 5 feats
group z-norm if K ≥ 3 · log-softmax within K
Actions
dispatch
(job, machine)
assign AGV
V1 / V2 / V3
maintain
NOW / DEFER
Two-phase training · one model · cross-instance
A · MA only
G rule samples per top-level point
group advantage
update: PA / LA / RA select heads
encoder shared
phase A
B · scoring heads only
rules frozen at rule*
M execution rollouts
group advantage
update: score_pa / la / ra
phase B
loss = −Σ Â · log p
no PPO clip · no critic
train: random MK-range instances
test: MK01–10 zero-shot
layout seeds disjoint
one shared encoder